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Where does the edge come from in quantitative terms?

Retail trade is coming

Evidence-based research report for a virtual AI and ML trading company

Research completed September 8, 2026 · Holding periods 15 minutes to several days

1 Executive summary

1. Strong for the fundamental separation: A statistical return pattern is only then an economic one

Edge, if it persists with information available at the time, executable orders and by cost;

reproducible anomalies and profitable strategies are different findings (Chen and Velikov 2023).

2. Strong as a theory, open as a concrete trading opportunity: information costs and limited arbitrage capital

explain why prices do not have to perfectly reflect all information at all times; they guarantee

no profit for any particular retail actor (Grossman and Stiglitz 1980; Shleifer and Vishny 1997).

3. Moderate for today's retail transfer: Historical evidence also exists directly for

15 minute ML forecasts, but current, independently replicated live evidence under the conditions

this company is missing (Aleti, Bollerslev and Siggaard 2025).

4. Strong historical, moderate in mechanism mapping: intraday momentum and short-term reversal

fit time-bound hedging demand, deferred processing and liquidity provision; theirs

Direction depends on the time window considered (Baltussen et al. 2021; Heston et al. 2010).

5. Moderate: ML improves the use of weak and interacting predictors in individual studies; the

The economic origin of the signal still lies in information, behavior, risks or trading frictions

(Gu, Kelly and Xiu 2020).

6. Strong: Costs and selection biases can erode perceived benefits, while a high

Model complexity alone does not provide any evidence gain (Schwarz et al. 2025; Bailey et al. 2017).

7. Own prioritized derivation: The company should initially have a few causally plausible, precisely defined ones

Check hypotheses for liquid instruments, protect costs and data times from model complexity and

then measure the incremental benefits of ML.

8. Overall verdict, moderate evidence: A positive retail edge in terms of costs is possible, for those here

unspecified company but not documented; neither a universally best horizon nor a serious target sharpen allows

derive from this literature.

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2 assumptions and terms

Standard assumptions. "Virtual Quantfirma" serves as a neutral company name. A separate one is being investigated

small to medium trading account without privileged infrastructure; "Virtual" describes the organization, not that

Quality of simulated fills. The focus is on liquid US stocks and broad ETFs because they are special for them

relevant original works are available. Futures, FX and crypto serve as a comparison; Options just appear

to classify the volatility premium. This is not an assumption about residency, broker license or

actual product availability.

Data. Public publications and regularly purchasable retail data are permitted. A public

Accessible data set is neither automatically free of charge nor of sufficient quality at the time

Fully available at this time. For intraday, historical quotes are required in addition to trades; pure

OHLCV candles are a possible forecast input, not a complete execution model. Exclusive

Order flows and institutional special conditions are excluded.

Target size. Expected net return, risk-adjusted additional return, drawdown and stability are evaluated

outside of the development samples. Taxes are not modeled due to the lack of legal form and jurisdiction.

Trading costs include spread, commissions, additional execution deviations, market influence, borrowing

and financing; Data, computing power and personnel must be borne separately as company costs. There won't be one

Capital amount and no return target invented.

Which means edge here. Operational edge is a positive expected return on an actually executable

Decision relative to a suitable alternative. Alpha denotes more closely the after specified

Risk factors remaining additional yield. A risk premium can be profitable without mispricing or

to be alpha. A cost saving is an improvement over the same decision made worse

execution; it alone does not generate positive strategy expectations.

3 Research methodology and standard of evidence

The research is a structured, qualitatively systematic evidence synthesis with documented search families

and targeted cross-checking. It is not an exhaustive full database survey or a meta-analysis.

The search was carried out on September 8, 2026 via web search, publisher websites, university and author repositories,

NBER, Fed, SEC and selected original contributions from practitioners. Published versions were opposite

older working papers preferred; pure reflections of the same paper do not count as replication.

The search families covered EMH/AMH and arbitrage limits; Intraday momentum, order flow and reversal; ML

asset pricing and daily statistical arbitrage; Multiple testing and backtest selection; replication,

publication decay and net costs; earnings drift; Carry, volatility premium and crypto fragmentation; public data and retail execution. Examples of actual search terms used include "Market intraday momentum", "The Price Impact of Order Book Events", "Empirical Asset Pricing via Machine Learning", "Replicating Anomalies", "Zeroing In on the Expected Returns of Anomalies", "Intraday Market Return Predictability Culled from the Factor Zoo" and "Implications of Transaction Costs... Post-Earnings Announcement Drift".

Original works with an identifiable object of study or theoretical ones were included
Mechanism and official information on data and routing availability. Sources without sufficient Verifiable attribution, return advertising and unverified new preprints were not considered supporting evidence used. The second round of searches aimed at opposing findings, differing replication definitions, and current ones Publication versions and the lack of a direct 15-minute reference. The search ended when everyone Response dimensions were documented or their gaps were explicitly marked; there would be no additional hits reliable general retail success rate delivered.

Animal is source type, evidence strength is claim quality. Tier 1 includes peer-reviewed articles and academic fundamentals; Tier 2 working papers and primary regulatory information; Tier 3 substantial

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practitioner research; Tier 4 forums. A single Tier 1 backtest remains in terms of today's tradability limited. An official product page proves a function offered, not its profitability.

Strong means well supported in the respective scope, for example by broader findings, direct Execution data or a clear theoretical derivation. Moderate means convincing individual findings or several findings with relevant transfer or identification limits. Weak means more inadequate direct evidence. Anecdotal refers to unverified self-reports. Recommendations and calculation examples are reported as their own derivations, not as empirically tested rules.

4 Theoretical basics

Efficiency and costly information

The EMH formulates information processing in prices; Tests must be the basis

Specify the amount of information and the model of normal expected returns. Weak, semi-strict and strict efficiency refers to different amounts of information. A predictable return

Therefore, efficiency does not automatically be refuted: it can contain a conditional risk premium. Strong as conceptual framework, not a universal proof of complete efficiency (Fama 1970).

If obtaining information costs, complete free disclosure in prices would provide the incentive to do so

Eliminate procurement. The Grossman and Stiglitz model therefore allows a limited

Information rent. Strong as a model result, but no evidence that publicly available OHLCV data is one unused pension included (Grossman and Stiglitz 1980). Arbitrage limits and adjustment

Mispricing can last longer if specialized arbitrageurs make losses, capital withdrawal and others

Having to endure misvaluation before convergence. "Wrongly priced" therefore does not mean risk-free

can be corrected in a timely manner. Strong as a theoretical mechanism; This results in the obligation for the company to Model liquidity and loss paths in addition to final returns (Shleifer and Vishny 1997).

The AMH views efficiency as the result of competition, adaptation and environmental conditions. From this it follows the plausible expectation of changing opportunities, not the identification of the next profitable regime.

Plausible thesis, but operationalized to a limited extent as a general forecasting theory; the original post presents a conceptual approach (Lo 2004).

Who pays for the benefit?

For every research project, a counterparty or economic function should be named: a hedger

paid for timely risk transfer; an investor reacts to information with a delay; a liquidity taker

paid for immediate execution; another market participant is limited by financing or mandate.

These questions are research heuristics in their own right. There is no substitute for a compelling story

Identification: The fact that a return coincides with stress does not yet prove whether risk compensation, Misvaluation or changed execution is its cause.

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5 Taxonomy of Edge Sources

The time information describes empirical windows or areas of investigation. "Not estimated" means no reliable universal signal half-life

is present. The classes overlap; ML can map multiple mechanisms at the same time.

Edge source

mechanism

horizon

Level of evidence and animal

Retail accessibility

Expected half-life
or decay
Main sources
information and
Processing
News correct and earlier
as a relevant comparison group
process
Event dependent;
Minutes to days
T1: theoretically strong; concrete
public retail pension weak
Public announcements
attainable;
Information advantage
prove separately
Not appreciated; Event time
and actual
Reception time is crucial
Grossman/Stiglitz
1980; Ng et al.
2008
Behavioral and
Earnings drift
Delayed response to
results; friction-related
slow adaptation
days until
weeks/months;
short neckline
test separately
T1: historical phenomenon
strong; explanation and
Net transfer moderate
reports accessible;
Surprise measure and
historical data quality
limit
No uniform
half-life; multi-day
Exit is its own hypothesis
Bernard/Thomas
1989; Ng et al.
2008
Structural intraday
Momentum
Time-bound hedging and
Rebalancing demand
Especially the last 30
minutes
T1: broader historical evidence
strong; Today's version open
Liquid ETFs/Futures
fundamentally researchable
closing window; later ones
Reversal possible, none
universal decay number
Gao et al. 2018;
Baltussen et al.
2021
Short-term reversal
and liquidity
Temporary price concessions,

Inventory and liquidity risk
 Under one
 hour and daily
 che/multi-day
 Constructions
 T1: mechanism and
 historical patterns strong;
 Retail net moderate
 Conditional; aggressive action
 can eat up the premium
 Depending on strategy; Intraday bounce
 of repentance lasting several days
 separate
 Heston et al.
 2010; Nagel 2012;
 de Groot et al.
 2012
 Momentum and value
 as risk styles
 Trend continuation or
 assessment differences; Risk
 and behavior as explanations
 Classic
 Evidence
 predominantly
 months and longer
 T1: for classical horizons;
 15 minute transfer weak
 instruments accessible;
 Horizon often doesn't fit
 None from monthly studies
 derivable
 Intraday half-life
 Moskowitz et al.
 2012; Jensen et
 al. 2023
 Carry and
 Volatility premium
 Yield for
 Financing/risk bearing or
 Insurance against volatility
 Rewards
 longer term
 appreciated;
 positions
 possible at short notice
 T1: premium receipts; more concrete
 1-5 day edge weak
 Conditional; derivatives, lending,
 Margin and tail risks
 No pure alpha decay;
 Premiums vary with
 states
 Kojen et al. 2018;
 Bakshi/Kapadia
 2003
 Order flow and spread
 CaptureImbalance in the order book,
 passive liquidity
 Seconds to
 minutes
 T1: price effect strong;
 independent

15 minute forecast weak
Quotes partly available for purchase; Cue
and realistic fills are missing
Candles
Cont does not measure a general one
prognostic half-life
Cont et al. 2014

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Edge source
mechanism
horizon
Level of evidence and animal
Retail accessibility
Expected half-life
or decay
Main sources
Execution
Spread, timing, turnover and
avoidable deviations
reduce
All horizons
T1: Cost impact strong; T3
supplementary
Choice of broker, order rules,
Fill measurement can be used
No signal decay; Savings
relative to
Comparison version
Black et al.
2025; de Groot et
al. 2012; Frazzini
et al. 2018
Statistical and ML
Regularize weak signals
and combine; Interactions
Direct studies on
15 minutes and
days, next to it
months
T1: single positive
OOS findings;
Retail transferability moderate
Models available;
Data needs and pipeline
demanding
To estimate empirically;
Model class does not return any value
before
Aletti et al. 2025;
Krauss et al. 2017;
Gu et al. 2020
Alternative data and
Fragmentation
Additional observations or
segmented markets
Data set and
market dependent
T1 for historical
crypto fragmentation;
generic alternative data pension
weak

Public data yes; history,
Licenses and capital transfer
limit
Not appreciated; Access and
Competition can be an advantage
eliminate
Makarov/Schoar
2020; SEC
EDGAR 2025

The short references are completely broken down in the bibliography, which is organized by tier. Retail accessibility is an assessment under standard assumptions, no examination of a specific broker account.

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6 Horizon Analysis

15 minutes

Direct positive finding, moderate retail transfer. Aleti, Bollerslev and Siggaard (2025) combine 272 high-frequency portfolio series and forecast 15-minute returns; OOS covers 2004-2020. The In the paper, SPY's main strategy achieves an annualized intraday sharpness of 1.37 according to its cost model: half quoted spread per position change and 2% annualized borrowing costs. The overnight position is on set to one, while the intraday metric excludes overnight returns. This is not a completely daily one closed retail account. Commissions, operational delays and data costs are not complete secured; the current independent retail replication remains open (Aleti et al. 2025).

Strong microstructure observation, weak prognosis transfer. Cont et al. explain price changes essential due to simultaneous order flow imbalance, best bid/ask; the base scale is ten seconds. This is not the same as predicting future 15 minute returns. A regression of price changes on the same interval order flow can be economically instructive and yet as Trading signal may be unusable (Cont et al. 2014).

Own implementation conclusions. For 15 minute candles, a signal must be calculated from the finished candle only be available after graduation. The closing price of the same candle is without explicit Auction/order model no guaranteed entry. Quotes, delay, time synchronization of the Predictors and minimum costs should be examined first. The 15 minute holding period alone also means not 26 round trips per US trading day: no-trade rules and positions held over multiple intervals can have a significant impact on sales.

Hours and clearly defined meeting windows

Gao et al. examine the forecast of the last half hour through the return from the previous close until the end of the first half hour. The overnight component is part of the signal. The historical one SPY findings 1993-2013 and other ETFs provide strong evidence for this specific pattern, not for any consecutive candles. A signal observed in the morning does not necessarily mean that the Position is held continuously until the end (Gao et al. 2018).

Baltussen et al. expand the evidence to 62 futures and link the late continuation with Hedging demand. Your signal extends from the previous day's close until 30 minutes before the close. A subsequent one Reversal over days matches temporary price pressure. Strong as a cross-market historical finding, moderate for full causal explanation (Baltussen et al. 2021).

Heston et al. document continuation in the same half-hour window after integer trading days as well as very short-term reversals. The return of at least 40 days back describes a seasonal layer structure; it is neither a 40-day holding period nor a half-life. Strongly historical, with significant Implementation reservations (Heston et al. 2010).

Own test idea: First use a clearly defined market and closing time. Then between Differentiate forecast time, entry time and exit and a predetermined postponement of the entry test. Transferring to FX or crypto markets needs its own definition of "session"; the US stock closing logic is not automatically present there.

One to several days

Short-term stock reversals can represent reward for providing liquidity. Nail connects hers expected returns with stress and withdrawal of other liquidity providers. Strong for the historical connection, moderate for mechanism identification; a retail liquidity taker can incur costs pay that the modeled provider earns (Nagel 2012).

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de Groot et al. show that large-cap selection and fewer position changes were modeled Can improve net reversal income. Results are based on historical US and European data

and institutionally influenced cost assumptions. Moderate as actionable evidence, strong on the importance of implementation; This does not result in a current return forecast (de Groot et al. 2012).

Earnings drift is a classic finding of delayed price reaction, but its usual examination is sufficient over weeks and longer. Costs are precisely related to the larger gross effects. For an exit

A separate event test is therefore necessary after one to five days. Strong historical, moderate to weak for this short net excerpt (Bernard and Thomas 1989; Ng et al. 2008).

Krauss et al. examine daily long-short forecasts in the S&P 500 universe and find in the historical

OOS test positive results as well as diminishing returns over time. This is a closer match

ML findings as monthly asset pricing, however, remain a backtest. Moderate; the exact one

Retail cost calibration is not fully verified in this research, so there is no net figure

adopted (Krauss et al. 2017).

What a half-life actually measures

A signal half-life, the typical holding period, the observation window width and the decay of a

published effect over calendar years are four different sizes. With an accepted one

exponential decay would be $a(\delta) = a(0) \times 2^{(-\delta/h)}$, where h is the half-life after a delay δ

remaining advantage. This is a model assumption, not a general market law. A signal

can only take effect at the end or change sign; then a single positive fits

Half-life does not.

Own measurement suggestion: With a frozen model, the conditional future returns, net income and

Estimate forecast accuracy over multiple predefined delays and hold times. Uncertainty arises

Consider daily/event level. A favorable 30-minute value does not occupy the neighboring one

15 minute value. Publication decline is analyzed separately in the calendar: McLean and Pontiff document in

Mean 26% lower returns outside the original sample and 58% after publication over 97

Predictors, no general half-life (McLean and Pontiff 2016; strong for historical mean

weakening).

7 What ML can contribute and how false signals arise

Performance and limits

ML offers regularization, flexible interactions, and efficient processing of many features. Gu et al. find in

monthly stock forecasts benefits for specific trees and nets; the investigation concerns

Risk premiums. Moderate for methodological added value, weak as direct intraday evidence (Gu et al. 2020).

The comparison "ML can do something that statistics fundamentally cannot do" is misleading: regularized

Regression, non-parametric estimation and state models are also part of statistics. What is crucial is

the reliable model comparison, not the name.

Own prioritized applications: estimate conditional return distributions instead of just direction; Signals

combine; Skip trades if there is insufficient distance to cost; volatility and

Model execution probability. Whether this improves the portfolio return remains to be tested separately.

Better volatility forecasts do not automatically mean better directional forecasts. Feature Importance or

SHAP values do not identify causality. An LLM can structure texts and support research; one

current generic LLM commercial pension has not been proven here.

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The main distortions

problem

Specific source of error

Necessary cross-check as

own validation suggestion

Look ahead

Revised balance sheet data; News spread later; Fill

at the already known candle close

Publication, reception and

store execution time separately;

Point-in-time features

Survivorship

Today's index members used retroactively; delisted titles

missing

Time-dependent universe, delistings

and reconstruct capital measures

Temporal leakage

Random splits with overlapping target returns;

Normalization to overall data

Chronological external tests;

all transformations only in

training window; overlapping
 Remove information intervals
 Multiple testing
 Lots of features, seeds, stops, assets and horizons; just
 Winner reported
 Log entire search history;
 Selection correction and final
 Use untouched test
 Execution bias
 Every limit touch as a fill; Midpoint as a tradable price;
 Shorts always available
 Conservative fill rules,
 Quote/order protocols,
 Non-executions and loans
 take into account
 Incorrect sample size
 Thousands of correlated trades as independent observations
 Dependency over days, markets and
 note events; robust or
 block-wise uncertainty estimation
 Regime and selection bias
 Only crisis phase or best universe selected
 Pre-defined partial periods,
 Stressful phases and negative controls
 report

The table operationalizes known selection issues for this firm; it is not empirical as a package
 proven recipe for success. Harvey et al. show why usual individual test hurdles in the factor zoo are too low
 can. Your oft-quoted t-hurdle above three is context-dependent and not a general release threshold for
 Intraday models (strongly methodological; Harvey, Liu and Zhu 2016).

Bailey et al. develop the probability of backtest overfitting as a diagnosis of strategy selection: it is not
 the likelihood of losing money in the future. The deflated Sharpe ratio takes into account selection and
 Non-normality under their assumptions. Both methods require plausible information about the actual
 search carried out and do not replace either correct data or later tests (strongly as methodological
 Warning, limited as a guarantee of protection; Bailey et al. 2017; Bailey and López de Prado 2014).

Why the replication literature seems contradictory

Hou et al. find under their economically changed construction with NYSE limits and value weighting for
 65% of 452 anomaly variants did not reach significance at the usual hurdle. Chen and Zimmerman
 In contrast, 158 of 161 clearly significant predictors reproduced in the original. The population and the
 Question differ: exact historical reconstruction on the one hand, robustness to trade and
 Weighting decisions on the other hand. Strongly in favor of dependency on test design, no evidence that
 either almost all papers are wrong or almost all strategies are profitable (Hou, Xue and Zhang 2020; Chen and
 Zimmermann 2022).

Jensen et al. are found in an international, thematically and Bayesian structured analysis
 Replication. Chen and Velikov, on the other hand, show how costs, publication timing and
 Selection corrections greatly reduce economic returns. These results can apply side by side.

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Moderate to strong within each construction, no consistent net return estimate for
 Retail (Jensen, Kelly and Pedersen 2023; Chen and Velikov 2023).

8 Retail reality check

Access is gradual

Co-location, own low-latency infrastructure and exclusive data are not included after the task has been accepted
 to the resource inventory. This does not result in a blanket ban on all institutionally used functions:

The official IBKR execution policy also describes SmartRouting for customer orders. Strong for them
 Function availability, without commitment to a specific account or product (IBKR execution policy).

PFOF is a payment and routing arrangement between intermediaries, not one from the end customer
 extractable alpha source. Broker-mediated execution on various trading venues does not have its own
 equivalent to direct venue membership. Black et al. also find that differences in PFOF are their
 do not explain the differences in execution observed. Strong in the US experiment examined, not as
 Statement on the current legal situation in all countries (Schwarz et al. 2025).

What costs destroy the edge

Own calculation identity: Expected net income per round trip = expected gross income – total

Spread contribution – commissions – additional slippage/impact – loans/financing. With an immediate purchase to the ask and later selling to the bid, two half spreads equal approximately one full spread, provided Quota and reference are consistently defined. Will the entire deviation from the Measured at the decision midpoint, the spread may not be added again.

Black et al. measure with around 85,000 simultaneous market orders in 128 US stocks from December 2021 to June 2022 median account-based round-trip execution costs of 7 to 46 basis points excluding Commissions. These are historical averages of specific stock orders and brokerage accounts, not today's averages Tariffs and no cost approach for SPY, futures or any retail account (strong and direct; Schwarz et al. 2025). Hypothetical sensitivity, not a market estimate. All numbers in the following table are freely placed Calculation scenarios in basis points of the traded position value. One basis point is equal to 0.01%. The Costs apply per round trip; "Remainder" here includes additional slippage and, if necessary, financing. This one Scenarios are not a prediction that longer trades will produce higher gross returns.

Example

Gross

spread

Commission

Rest

Net

15 minutes, cheap version

3

1

1

1

0

15 minutes, higher friction

3

3

2

2

–4

hours, low buffer

8

2

2

2

2

days, larger buffer

25

3

2

5

15

Same day trade under cost stress

25

8

4

15

–2

Sales calculation as a separate illustration: Four complete round trips per day, each with 25% of the account value and 5 bp costs, consume $4 \times 0.25 \times 5 = 5$ bp account return daily. Over an assumed 250 In days, this is arithmetically 12.5% of the assumed constant account value, without compound interest effect. Same Cost per trade can destroy a high-frequency traded weak forecast without a rare traded stronger forecast to destroy. The horizon therefore affects signal size, turnover and Financing, not a fixed fee rule.

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Small accounts and capacity

Two limits can be distinguished: upper market capacity due to market influence and available liquidity as well as lower economic account size due to fixed costs, minimum order sizes and lack of Diversification. Small orders can cause less impact; small accounts carry data and Operating costs but on less capital. A weak small account backtest doesn't prove an upper one Capacity limit.

Own illustrative calculation: With hypothetical fixed research/data costs of 100 euros per month 10,000 euros of capital must initially be 12% annually and 100,000 euros of capital must be 1.2% annually before these Earn fixed costs just to cover them. One euro minimum commission per page costs 500 euros Position value alone 40 bp per round trip, for 5,000 euros 4 bp. These are not broker prices. Nominal Position size, account capital and leverage should not be confused in this calculation. Frazzini et al. examine real institutional trading data and show the dependence on costs and Impact of trading circumstances. Their scale and execution capabilities are none Retail conditions (Tier 3, methodologically substantial; Frazzini, Israel and Moskowitz 2018). Are for the company a separate cost curve by instrument, time of day and order size as well as separate measurement of Opportunity costs of unexecuted orders required.

Data and asset classes

The SEC offers publicly available APIs for filing history and XBRL data without an API key. Data access is This proves that there is no information advantage. A today's retrieval must be based on that time for historical tests submitted versions and their publication date can be traced back (official Primary Information, Tier 2; SEC EDGAR APIs).

Own procurement priority: First, company and instrument identifiers, corporate actions, Secure delisted titles, quote/trade timestamps and your own fills. Then add event-related texts.

Only buy alternative data if it closes a specific information gap and is unadulterated History and usage rights are available. For indeterminate satellite, web or sentiment data, go here no proven universal retail benefit.

Stocks/ETFs offer appropriate literature, but require clean universes and, if necessary, borrowing. Futures require contract, roll and margin treatment in their own implementation; synthetic Updated time series are not directly tradable prices. FX results are of the concrete Spot/futures/broker instrument dependent. These points are implementation requirements, not Statement about current conditions.

Makarov and Schoar document historical price differences between crypto exchanges and the significance of capital movement limits. Fragmentation can exist precisely because the Arbitrage capital transfer is limited. Strongly historical, no evidence that it can be skimmed off free of charge today Differences (Makarov and Schoar 2020). If you do your own test, transfer time would be available at the same time Capital, default risk and, if necessary, funding must be recorded separately.

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9 Negative analysis and vulnerable assumptions

Claim

Judgment and justification

Document situation

A more complex network is created

Edge

Not documented. Additional degrees of freedom also increase the search space; a historical model gain is not a universal one

Advantage of deep networks.

Moderate; Gu et al. 2020, Krauss et al.

2017; Selection risk Bailey et al. 2017

Provide standard indicators

reliable net alpha

For the DJIA rules examined, see Bajgrowicz/Scaillet

lack of ex ante selection and cost sensitive

Results. No evidence against any conceivable price-based

Strategy.

Strong within the tested daily

rules; Bajgrowicz/Scaillet 2012

Every reversal curve is a

exploitable pricing error

Bid-ask bounce and liquidity risk can have the effect

explain; Anyone who acts aggressively deserves to be observed

Don't spread.

Strongly mechanistic; Heston et al.

2010, Nagel 2012

Monthly momentum or

Value confirmed

Intraday trading

Incorrect horizon transfer. Monthly predictors and

Portfolio building doesn't validate a 15-minute decision.
 Strong for the design difference;
 Moskowitz et al. 2012, Gu et al. 2020
 Short Vol or Carry is
 risk-free alpha
 Premiums can compensate for risks. A short one
 Hold date does not remove risk during the position.
 Moderate to strong; Bakshi/Kapadia
 2003, Kojen et al. 2018
 Zero commission means
 free of charge
 Execution still incurs costs; real
 Account results vary.
 Strongly direct; Black et al. 2025
 More backtests or one
 nice walk-forward
 demonstrate robustness
 Repeated selection after visible test results makes
 these to development data. Strongly methodical; Harvey et al. 2016,
 Bailey et al. 2017
 Experience and persistence
 guarantee success
 Historical day trader data only shows a small amount reliably
 successful group; more activity alone does not demonstrate learning.
 Strongly historical, transfer limited;
 Barber et al. 2014
 Bajgrowicz and Scaillet examine 7,846 technical rules on daily DJIA data from 1897 to 2011. Already
 Low modeled transaction costs eliminate the performance under consideration. Strong negative finding for
 these rules and this market environment, not a blanket refutation of all technical or
 intraday-based models (Bajgrowicz and Scaillet 2012).
 Barber et al. For Taiwan 1992-2006 find less than 1% day traders with predictable and reliable
 positive abnormal returns after fees. That doesn't mean that exactly 99% lose every year or
 that today's AI retail companies have the same rate (strongly historical, external validity limited; Barber et al.
 2014).
 Attackable prerequisites of the key question. First, Edge may be stateful instead
 permanently. Secondly, ML, strategy and company do not form identical success factors: a positive strategy
 can be unattractive after fixed costs. Third, "retail" is not a uniform cost profile. Fourth, they close
 desired horizons some well-documented rewards. Fifth, a high backtest can sharpen
 reflect hidden crisis exposure, selection or unrealistic prices. Sixth, one can completely
 simulated company cannot prove its execution quality solely from its own simulation model. These are
 own critical conclusions from the analyzed measurement and transfer problems.
 10 Implications for the Virtual Quant Firm
 The following order is a separate research and implementation prioritization. She claims none
 already tested ranking of the most profitable strategies. Each phase requires a verifiable result; the
 next step depends on the result.

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Priority 1 Set data and economic target

Choose a small liquid universe, broker/instrument type and time zone rule.
 Corporate actions, membership at the time, trading halts and shorts availability
 hold on. Save decision midpoint, order time and fill price separately. The primary goal is net additional income
 relative to a fair, simple alternative with comparable risk; daily account returns, drawdown and
 Tail losses complement the Sharpe. Annual projections cannot ignore autocorrelation.
 Verifiable result: A data set with the provenance and actual availability of each feature; one
 deterministic cost accounting; a reproducible basic baking test without ML. If one of the sizes is missing,
 the signal has not yet been assessed as tradable.

Priority 2 Three limited research projects

A — Session-bound continuation. The exact Gao/Baltussen time definition is initially faithful to the original
 replicate with a suitable liquid instrument. Then select a predetermined newer period and costs
 use your own version. A freely chosen "intraday momentum" is not a replica.
 B — Daily reversal or event hypothesis. With clearly limited sales, check whether after
 observed price/information event there is still a net buffer. Earnings signals require one

comprehensible level of surprise; a mere positive report is not a defined surprise.

Long-only and long-short are separate strategies.

C — Direct 15 minute ML replication. The original construction described in Chapter 6 including Reconstruct data requirements, costs and overnight treatment. Only then does the deviation “small Retail company with reduced features” as your own experiment. Less data and different execution are no meaningless implementation details.

Verifiable result per project: An economic hypothesis, a frozen set of rules, one complete list of all experiments, an original replication test and a separate later test.

The projects serve to generate evidence; they do not justify automatic capital investment.

Priority 3 Check ML against strong simple baselines

Compare regularized linear models and simple rule-based variants first; then a limited one Tree model and only a network for a specific additional hypothesis. Same data, costs, risk budgets and Use tests. Remove groups of characteristics and check whether the additional net income is stable or just Result of higher market/volatility exposure. A no-action model is a serious one Alternative.

Priority 4 Use a mandatory validation path

1. Train chronologically and separate inner model selection from outer OOS test; overlapping labels and do not spread related events across borders.

2. Search history including rejected ideas, universe changes, and cost assumptions document; Use selection corrections as supplementary diagnostics.

3. Leave a final, more recent period untouched; one-time evaluation with predetermined ones Key figures and economically justified cost stress.

4. Observe live data initially without orders to check reception times and signal reproducibility; Mark paper fills as assumptions.

5. Only measure real execution on a small scale when the criteria have been met and the loss budget has been set; then check live deviations against the expected cost and risk range.

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Termination criteria: The benefit disappears if there is a plausible delay or cost; focuses on not tradable securities; requires repeated adjustments to the test; or cannot be statistically separated from zero as soon as temporal dependency and selection must be taken into account. A justified task of a hypothesis is a successful research result.

Priority 5 Manage expectations and operations

Do not derive monthly profit promises from historical studies. Model, data and versioning execution changes; Limit violations, missing quotes and timestamp deviations must be able to stop trading. A strategy only within measured liquidity and cost ranges scale. The company is only economically plausible if there is a resilient net buffer as well as the fixed ones bears operating costs.

11 What practice believes and science has not yet proven

The following forum threads are specifically selected examples, not a representative survey Practitioner consensuses. Your statements are expressly unverified practical anecdotes (Reddit, Tier 4).

In the thread “Where does ML fit in algorithmic trading?” ML is used as a filter, regime tool and Execution help preferred over a general alpha finder. That’s a plausible one

Working hypothesis, but the claimed general superiority of filters is not systematic

Retail comparison proven. The primary literature both justifies and contains consideration of incremental benefits direct return forecast successes; an absolute “ML can only filter” would be too strong (Reddit thread; comparison: Gu et al. 2020, Krauss et al. 2017).

“Trading using ML” shows the typical controversy between own cost/leakage insurance and the accusation of overfitting. Both remain without complete data, experimental history and independent Execution measurement unchecked. Methodological research supports concerns about selection bias, but may neither confirm nor refute the poster’s specific strategy (Reddit thread; comparison: Bailey et al. 2017, Harvey et al. 2016).

Success screenshots, the number of indicators used and the self-description as a quant do not provide any reliable evidence. Conversely, “retail can basically never win” is also stronger than the historical one Research allowed. The sensible limit is between auditable net income and unaudited net income Claims.

12 research gaps

Current retail transfer: This corpus lacks an independent, account-specific live replication of the prioritized models up to the research day. A 2025 release may use data that years end beforehand. The answer to “does it work today?” therefore requires new observations.

Decay under feasible conditions: Curves of the remaining advantage are missing for the specific company

according to actual data and order delay. The decomposition into signal decay, spread change, Adverse selection of fills and calendar regimes is open.

Data economy: It is unclear whether a reduced public feature base will make up for the sometimes extensive academic factor data can replace and whether remaining profits cover data and infrastructure costs cover. Model accessibility alone does not solve this question.

Mechanisms: Liquidity, tail risk and behavioral interpretations are often not clearly separated.

Countertests should check whether a model systematically loses in poor conditions and exploits stale prices or just reproduces a known factor exposure.

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Market and regime transfer: US stock findings cannot be generalized to European execution, FX brokers or crypto exchanges. For crypto and FX, the research was less broad; the same No depth of evidence is claimed.

13 Uncertainties and limitations of this research

The research verifies sources and relevant statements, but does not carry out its own data replication, Meta-regression or live strategy check. It includes 34 cited or official sources

Documents/Threads in the following directory. The selection is problem-oriented, not random complete sample of all financial studies. Evidence strength is not calculated from hit numbers.

Some publishers blocked direct full-text retrieval. Original abstracts, indexed original sections or Author versions were then used. The bibliography states the depth of access. Exact

Cost calibrations, particularly at Gao, Krauss, and Nagel, have not been fully examined; become of it no unsecured net return figures are assumed. The direct 15 minute work was published in the Read the version including the relevant cost and overnight passage.

Publication bias, market changes over time, shared data sets and possible model dependency also limit Tier 1 evidence. Theoretical models explain possibilities under assumptions, none Return guarantee. Reproduction rates, historical success rates and modeled Sharpe ratios are therefore not translated into probabilities for the success of this company.

Without a specific account, universe, data plan and trading protocol, neither current total costs nor a target capacity or target sharpness can be reliably specified. Tax and legal issues as well as today Availability of specific instruments in the country of residence was not investigated. The cost examples are only transparent calculation assumptions.

The viable answer therefore remains precisely limited: The possible edge lies in specific information and Liquidity frictions, risk bearing and their economically better use; whether the virtual quantum company can benefit from this at their own cost is an open empirical question.

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14 Bibliography by Tiers

All links were last checked for this research on September 8, 2026 or used as original evidence. "Full text" refers to the examination of relevant passages, not a claim of complete replication of all tables. The Titles below also serve as a clear breakdown of the short documents in the tables.

Tier 1 Peer-reviewed research and academic foundations

Aleti, Saketh; Bollerslev, Tim; Siggaard, Mathias (2025). Intraday Market Return Predictability Culled from the Factor Zoo. *Management Science* 71(9), 7731–7751. Study type: empirical ML-OOS backtest. Access: published Authors' full text, especially data, section 5 and cost treatment. DOI 10.1287/mnsc.2023.01657.

Bailey, David H.; Borwein, Jonathan M.; López de Prado, Marcos; Zhu, Qiji Jim (2017). The probability of backtest overfitting. *Journal of Computational Finance* 20(4), 39–69. Study type: methods/simulation work. Access: institutional evidence and author registration. DOI 10.21314/jcf.2016.322; Author version.

Bailey, David H.; López de Prado, Marcos (2014). The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting and Non-Normality. *Journal of Portfolio Management*, 40th Anniversary Special Issue, 94–107. Study type: Method/simulation work. Access: Authors' full text; DOI not verified here. Original version.

Bajgrowicz, Pierre; Scaillet, Olivier (2012). Technical trading revisited: False discoveries, persistence tests, and transaction costs. *Journal of Financial Economics* 106(3), 473–491. Study type: empirical new investigation of technical Rules. Access: Author's full text and publisher's abstract. DOI 10.1016/j.jfineco.2012.06.001.

Bakshi, Gurdip; Kapadia, Nikunj (2003). Delta-Hedged Gains and the Negative Market Volatility Risk Premium. *Review of Financial Studies* 16(2), 527–566. Study type: theoretical-empirical options study. Access: published Authors full text. DOI 10.1093/rfs/hhg002.

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Chen, Andrew Y.; Zimmerman, Tom (2022). Open Source Cross-Sectional Asset Pricing. *Critical Finance Review* 11(2), 207–264. Study Type: Reproduction and Open Research Data. Access: Fed full text 2021 and author code with proof of publication; Reproduction numbers from a verified version. DOI 10.1561/104.00000112; Fed version.

Cont, Rama; Kukanov, Arseniy; Stoikov, Sasha (2014). The Price Impact of Order Book Events. *Journal of Finance Econometrics* 12(1), 47–88. Study type: empirical microstructural analysis. Access: Authors' full text. DOI 10.1093/jffinec/nbt003; arXiv 1011.6402.

de Groot, Wilma; Huij, Joop; Zhou, Weili (2012). Another look at trading costs and short-term reversal profits. *Journal of Banking & Finance* 36(2), 371–382. Study type: historical strategy/cost baking test. Access: Authors' full text via University. DOI 10.1016/j.jbankfin.2011.07.015.

Fama, Eugene F. (1970). Efficient Capital Markets: A Review of Theory and Empirical Work. *Journal of Finance* 25(2), 383–417. Study type: theoretical and empirical basic review. Access: published original text. Full text. Note: Some publisher metadata also lists the meeting chair; The author of the article is Eugene F. Fama.

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Lo, Andrew W. (2004). The Adaptive Markets Hypothesis: Market Efficiency from an Evolutionary Perspective. *Journal of Portfolio Management* 30, 30th Anniversary Issue, 15–29. Study type: conceptual theoretical contribution. Access: Authors full text. Original text.

Makarov, Igor; Schoar, Antoinette (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Finance Economics* 135(2), 293–319. Study type: empirical market fragmentation analysis. Access: Original abstract and Proof of publication on MIT author page. MIT original credit.

McLean, R. David; Pontiff, Jeffrey (2016). Does Academic Research Destroy Stock Return Predictability? *Journal of Finance* 71(1), 5–32. Study type: empirical investigation of publication/OOS decline. Access: Publisher's abstract. DOI 10.1111/jofi.12365.

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Tier 3 Practitioner Research and Provider Disclosures

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Tier 4 Hypothesis Generating Forum Examples

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